Byung-cheol Kang

Genomics · Bacteriophages · Research Pipelines

A bioinformatics researcher and developer connecting biological questions with analysis design and interpretation. My work extends from microbes to humans, integrating diverse analytical approaches and building research pipelines and databases. AI-powered coding accelerates my research and amplifies my ability to automate workflows and develop analytical tools.

CURRENT AND RECENT ROLES

MicrobiotiX · Senior Researcher · Bioinformatics — 2026.07 - PRESENT

- Analyze bacterial and phage genomes and design host-recognition protein candidates.
- Develop phage research databases and prediction tools; contribute to cocktail selection software and provide analysis technical support.

Wittgenbio · Computational Biology Research — 2026.07 - 2026.10

- Built a CRC single-cell atlas, annotated cell types and analyzed expression of target candidates.

Past platform evidence. The integrated workflow and operational results in this portfolio describe work performed at Sanigen (2020.03 - 2025.11). They are not presented as MicrobiotiX or Wittgenbio assets or current-company outcomes.

PROJECT 01 · PLATFORM

Integrated Microbial Genomics Platform

PROBLEM

Separate assay scripts, environments, quality checks, server paths, and reporting conventions made analysis difficult to reproduce and review consistently.

SYSTEM DESIGN

Entry point	One Snakemake dispatcher selects 16S, ITS, WGS, or Hybrid profiles.
Execution	Server-specific YAML configuration with Conda, Docker, or Apptainer and pinned BioContainers.
Quality	Input validation, raw-read QC, assembly/analysis checks, review gates, and retained logs.
Delivery	Custom Python/R processing, visualizations, HTML reports, and packaged outputs.

ACTIVE TOOL CHAINS

Profile	Core processing	QC / interpretation
16S / ITS	QIIME 2, DADA2, SILVA classifier, MAFFT, FastTree	FastQC, MultiQC, diversity metrics, custom R/Python reports
WGS	Trimmomatic, SPAdes, Bowtie2, samtools, Prokka	QUAST, BUSCO, Kraken2, Abricate/ CARD/VFDB, eggNOG-mapper
Hybrid	NanoFilt, NanoPlot, Flye or Unicycler, BWA, Pilon, minimap2	QUAST, BUSCO, Prokka, CGView, BLASTn, Abricate, eggNOG-mapper

IMPLEMENTATION AND TECHNICAL REVIEW

- Connected shared analysis, container and server settings with QC, logs and HTML report outputs across the four execution profiles.
- Delivered WGS/hybrid assemblies under research-specific read-selection and trimming conditions, with reviewed contigs and reports.
- Performed assembly/Prokka annotation and manual contig review, with circular-genome, assembly-graph and BUSCO checks.

EVIDENCE BOUNDARY

Only tools connected to active Snakemake includes and rules are listed as active. Configured or documented but inactive components are not represented as production use.

PROJECT 02 · OPERATIONS

Analysis Operations and Quality Assurance

OPERATING MODEL

Question biological scope → **Execution** workflow + QC → **Delivery** interpretation + provenance

RESPONSIBILITIES AT SANIGEN

- Managed NGS analysis services and the bioinformatics team; performed genomics, microbiome and transcriptome analyses.
- Automated analysis workflows and reports and contributed to marker and panel research.
- Provided technical support to clients and internal researchers and operated analysis infrastructure.
- Contributed to the technology-special listing task force and prepared technical-evaluation and listing-review materials.

ANALYSIS DELIVERY AND AUTOMATION

- Delivered genome assemblies, annotations, strain comparisons, microbiome reports and transcriptome count/coverage outputs for research questions.
- Developed shared analysis configurations, QC and HTML reporting; automated result conversion and functional-annotation outputs and maintained phylogenetic/visual reports.

INTERNAL SERVICE OPERATIONS AND MANAGEMENT

- Managed analysis assignments, schedules, result review, SOPs, turnaround and quality.
- Handled client and internal-researcher inquiries about interpretation, additional analyses and reanalysis; coordinated the work schedules.
- Investigated and addressed analysis-environment, calculation and reporting issues; redelivered results.
- Reviewed NCBI chromosome/contig submissions and prepared analysis/panX training and handover materials.

ADDITIONAL CUSTOM ANALYSES

- Filtered vector/mixed-virus reads, assembled viral reads with metaSPAdes and identified assemblies by BLAST; delivered predicted prophage intervals and mobility evidence.
- Delivered OTU/sequence-match tables and hybrid-contig VCF/IGV comparisons with NCBI submission-related technical support.
- Verified allergen/accession mappings and custom BLAST target/toxin searches; reconciled NCBI/SILVA/species-label discrepancies with sequence/ANI evidence and documented marker limits.

TRACEABILITY AND INTERPRETATION

Retained communications and analysis outputs support these examples, without a complete workload total. Gene-count delivery is separate from completed differential-expression or enrichment analysis. Platform evaluation and RNA QC do not establish regulatory qualification or therapeutic efficacy.

From Genome Differences to Assay Candidates

COMPARATIVE-GENOMICS MARKER DISCOVERY

Objective. Identify a region that distinguishes a target group while remaining practical for downstream marker or assay development.

Approach. Curate complete genomes, compare whole-genome and pan-genome context, narrow candidate intervals, and review conservation, specificity, and downstream constraints.

Outcome. Prioritized an approximately **17 kb candidate region** for follow-up, with traceable reasoning from genome selection to practical assay constraints.

FOODBORNE-PATHOGEN NGS PANEL

- Automated k-mer and comparative-genomics searches for marker and primer candidates; prepared candidate sequences and comparison tables. Reviewed panel-analysis code and QA/QC.
- Contributed NGS data analysis and interpretation to collaborative panel development and evaluation.
- Supported interpretation of NGS outputs across food and environmental sample contexts.
- Collaborative outcomes included two peer-reviewed papers in 2023 and one joint patent contribution in 2022.

TECHNICAL-LISTING DOCUMENT CONTRIBUTIONS

- Reviewed diagnostic marker-search/panel functions, panX applicability and algorithm differentiation for technical and patent-review materials.
- Authored, revised and submitted technical-evaluation explanations, figures and tables; wrote the assigned preliminary-listing application section.

BOUNDARIES

> Computational prioritization does not establish diagnostic sensitivity or specificity. Publication and patent records document collaborative outcomes, not sole authorship or universal performance across unevaluated matrices.

SELECTED EVIDENCE

- **Frontiers in Microbiology** 14 (2023): agricultural wastewater pathogen panel.
- **Journal of Microbiology and Biotechnology** 33(1) (2023): fermented-food pathogen panel.
- Joint patent contribution (2022): NGS-based primer set and foodborne-bacteria detection method.

PROJECT 05 · CURRENT ROLE CASE STUDY

Klebsiella Phage Genomics and Tail-RBP Design

Reference sizes and benchmark rates on this case-study page document its input and evaluation context.

GENOME EVIDENCE WORKFLOW

Host + phage genomes
QC + annotation

→

Independent evidence
taxonomy + lifestyle + termini

→

Tail interpretation
KL target + structure

ANALYSIS DELIVERY

- Analyzed bacterial and phage genome quality, taxonomy, lifestyle and gene content; prepared phage candidate-assessment reports.
- Compared related phage genomes and organized references accounting for genome start positions and sequence orientation.
- Selected RBP candidates for target capsule types, generated domain-combination sequences and compared sequences and protein structures.

SOFTWARE AND DATABASE DEVELOPMENT

- Built an RBP reference database from literature and public sequences and implemented candidate searches by target capsule type.
- Developed a capsule-type prediction library and trained and cross-validated ML classifiers for enzyme–target pairs.
- Contributed to phage cocktail selection software based on measured host-range tables; verified target-strain and required-phage calculations for candidate combinations.

TECHNICAL SUPPORT

- Provided technical support for input, calculation and result-output issues in host-range, clustering and gene-analysis software.
- Prepared analysis specifications and reproduction records for external developer integration; compared execution results from source and installed packages.

TECHNICAL CASE EVIDENCE

The checked reference snapshot has 7,759 sequences in 5,467 clusters; K labels cover 442 rows and 62 labels, including 107 literature-backed rows. KL tables contain 446 part rows, 627 ranked entries and 800 evidence rows. Host-range input covers 221 strains and 74 phages. Saved 110-query-per-mode label inclusion was 85/110 (77.3%) excluding the query and 48/110 (43.6%) excluding its cluster.

ANONYMIZED REPORT CASE

- Four genomes were consistent with lytic **Przondovirus** calls and short DTR packaging with 180-183 bp repeats; three capsule targets matched supported references at 95.4-98.8% identity.
- A candidate's 129-aa anchor alignment covered 99.2% of the region; domain/structure evidence supported a module-replacement hypothesis for experimental follow-up.

INTERPRETATION BOUNDARY

The saved benchmark has 110 queries per mode and measures inclusion of at least one true target label, not complete label-set agreement or external validity. Its record correspondence to the current 107 literature-backed K-label rows is not established. Candidate sequences and structures do not establish adsorption, host range or lytic efficacy without experimental validation.

PROJECT 06 · WITTGENBIO RESEARCH

CRC Atlas and Cell-Type Annotation

CRC SINGLE-CELL ATLAS WORKFLOW

Single-cell datasets
QC + metadata

→

Integrated CRC atlas
harmonization + clustering

→

Cell-type annotation
cell labels + marker summaries

PUBLIC-STUDY INTEGRATION AND CELL-TYPE ANNOTATION

- Standardized expression matrices and metadata from public studies; performed QC, cross-study integration and clustering and prepared H5AD data.
- Assigned cell-type labels using cluster-marker expression and author annotations; produced cell-level annotations and marker summaries.

TUMOR AND NORMAL-TISSUE EXPRESSION ANALYSIS

CRC atlas
tumor expression

→

CRC candidate analysis
patient / specimen comparison

←

External normal atlas
tissue expression filters

- Annotated cell types and analyzed expression in an externally supplied normal-tissue atlas; compared tumor and patient/specimen expression and prepared CRC candidate tables using normal-tissue expression filters.
- Produced expression comparisons by cell type, tissue and patient/specimen, with CRC candidate analysis tables.

PUBLIC EVIDENCE BOUNDARY

These analyses support RNA-level candidate prioritization. Transcript expression does not establish cell-surface protein abundance, clinical safety, or treatment response. Private datasets and internal identifiers are not disclosed.

PROJECT 07 · INDEPENDENT R&D PROTOTYPE

Phage Lifestyle Evidence Prototype

RESEARCH QUESTION

Can existing tools be combined into an end-to-end system that distinguishes lytic and lysogenic phages while preserving conflicting evidence and uncertainty?

EVIDENCE DESIGN

Quality context	CheckV completeness and contamination context
Annotation	Pharokka and Phold; integrase, repressor, excision, and lysis evidence
Classifiers	BACPHLIP and PhaTYP as independent prediction evidence
Decision layer	Agreement/confidence table with unresolved cases routed to manual review

HOST-RANGE EXTENSION

A related hypothesis-ranking layer considers LPS and outer-membrane gene variation together with phage receptor and annotation evidence. It prioritizes candidates for experiment; it does not prove adsorption, infection, or lytic activity.

PUBLICATION-GRADE NEXT STEPS

- External, leakage-controlled benchmark with known lifestyle labels.
- Comparison against individual tools and transparent simple baselines.
- Sensitivity, specificity, MCC, calibration, and quality-stratified performance.
- Ablation analysis for gene markers, classifiers, and quality context.
- Experimental validation for host-range candidate claims.

CONTACT

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